FactoryFlow digital twin deployment challenges

"FactoryFlow turns CCTV into a Digital Twin: YOLO detects people/forklifts, ByteTrack tracks per camera, and homography maps positions to a floor plan. A 3D dashboard shows proximity alerts, near-misses, heatmaps and simulations. Find gaps blocking real-world deployment using recent research. Assess localization, calibration drift, cross-camera identity, safety validation, reliability, latency, privacy and scale. Rank evidence-backed gaps and fixes; separate proven gaps from assumptions."
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Deployment Gaps in CCTV-to-Digital-Twin Factory Safety Systems at a Glance

Key Findings Table

Rank	Evidence-backed gap	What the literature shows	Practical implication for FactoryFlow	Proven vs assumption	Key citations
1	**Safety validation is incomplete**	Studies report partial alert metrics, but the corpus rarely shows precision, recall, false alarms, delay, operator response, and longitudinal safety outcomes in one deployment; no retrieved factory/warehouse trial links camera analytics to reduced injuries over time [1] [2] [3] [4] [5].	A working detector is not yet a validated safety system; deployment risk is primarily evidentiary, not just algorithmic [6] [7].	**Proven gap**	[1] [4] [5]
2	**Calibration drift and maintenance are real operational risks**	Fixed-camera calibration degrades from view shifts, temperature, vibration, impacts, relocation, and scene change; multiple papers recommend online monitoring and self-calibration [8] [9] [10] [11] [12].	Floor-plan coordinates can silently become wrong unless calibration health is monitored continuously [13] [14].	**Proven gap**	[8] [10] [13]
3	**Cross-camera identity persistence remains fragile**	Geometry plus Re-ID improves multi-camera tracking, but failures remain under occlusion, similar clothing, lighting change, viewpoint shift, and fragmented trajectories; online methods are structurally weaker because they lack future frames [15] [16] [17] [18] [19].	Near-miss chains and heatmaps across blind spots will be unreliable unless identity stitching is upgraded beyond per-camera ByteTrack [19] [20].	**Proven gap for people; forklift case still weakly evidenced**	[15] [21] [22]

Rank	Evidence-backed gap	What the literature shows	Practical implication for FactoryFlow	Proven vs assumption	Key citations
4	**Localization is usable for zones, not universally validated for precise twin coordinates**	Reported projected-position errors range from about 13.2 cm to up to 1 m, with error increasing with distance and occlusion; most studies report aggregate accuracy, not factory-specific error decompositions by height, tilt, and forklift motion [23] [24] [25] [26] [27].	FactoryFlow is more defensible as a conservative zoning tool than as a precise localization layer [28] [29].	**Proven gap; precise-factory-performance claim remains unproven**	[23] [25] [29]
5	**False alarms threaten trust and compliance**	Across safety-critical domains, nuisance alarms produce desensitization, slower response, and ignored alarms; persistence and staged thresholds reduce alarm burden [30] [31] [32] [33] [34].	Frame-level proximity alerts are unlikely to be acceptable in real plants without alarm governance logic [35] [36].	**Proven gap**	[30] [31] [34]
6	**Latency/reliability evidence is thinner than the architecture claims**	Edge/fog/hybrid DTs are repeatedly recommended for latency and resilience, but many-camera CCTV benchmarks for alert freshness, frame drops, overload, and network loss are scarce [37] [38] [39] [40] [41].	Real-time safety use requires explicit service-level testing, not just edge deployment [42] [43].	**Partly proven gap**	[40] [41] [43]
7	**Privacy and governance requirements are clear, implementations are not**	The corpus converges on privacy-by-design, transparency, limited retention, role-based access, and auditability, but lacks complete factory CCTV implementations showing these controls end to end [44] [45] [46] [47].	Governance can block deployment even when models work [48] [49].	**Proven governance gap**	[45] [46] [47]
8	**Scale-out is constrained by integration and lifecycle operations**	Reviews repeatedly identify interoperability, orchestration, drift management, data quality, network load, and heterogeneous hardware as deployment bottlenecks; measured large-scale camera-fleet evidence is sparse [50] [51] [52] [53] [54].	Scaling from pilot to plant or multi-site rollout is a systems-engineering problem, not only a model problem [55] [56].	**Proven systems gap; exact scale limits remain uncertain**	[51] [53] [54]

Direct Answer

FactoryFlow is **technically plausible but not yet well de-risked for real-world deployment**. The strongest evidence-backed blockers are: **(1) incomplete safety validation, (2) calibration drift, (3) fragile cross-camera identity, (4) localization error for precise mapping, and (5) false-alarm management** [4] [8] [15] [25] [31].

A practical ranking of fixes is:

1. **Validate alarms as a safety system, not only a vision system**: report FAR, missed alarm rate, average alarm delay, response timing, and supervisor-verified outcomes; use staged and persistence-based alerts [1] [6] [7] [34].
2. **Add calibration-health monitoring**: detect homography drift online and trigger self-calibration or downgrade a camera to advisory-only mode [8] [11] [13].
3. **Upgrade cross-camera association**: use geometry in overlaps, Re-ID plus spatio-temporal gating across blind spots, and selective re-evaluation of risky matches [15] [18] [19] [20].
4. **Position the product first as a safety-zoning system** with conservative buffers, not as precise indoor localization [28] [29].
5. **Adopt edge-first fail-safe runtime testing** for overload, network loss, and alert freshness [40] [41] [42].
6. **Implement privacy-by-design and auditable governance** before rollout [44] [45] [46].

Proven gaps: calibration drift, localization degradation with distance/occlusion, cross-camera ID fragility for people, false-alarm consequences, limited large-scale latency evidence, and incomplete governance operationalization [9] [21] [26] [30] [43] [47].

Assumptions or weakly evidenced concerns: forklift-specific cross-camera Re-ID performance, exact recalibration frequency for factory CCTV fleets, many-camera dashboard freshness limits, and actual injury reduction from camera-based near-miss analytics in factories [4] [22] [43] [57].

Confidence level: High - the ranking aligns with repeated cross-source patterns, though some fixes are better evidenced than quantified in factory-specific trials [4] [10] [15] [51].

Study Scope

This review covers recent work, mostly **2020-2025**, across industrial vision, multi-camera tracking, proximity warning, digital twins, edge/fog/cloud architectures, and workplace privacy/governance [1] [20] [27] [37] [46]. It includes lab studies, field/pilot studies, benchmark-based tracking work, digital-twin architecture papers, and governance reviews [5] [19] [23] [54] [58].

Assumptions & Limitations

The evidence base is strongest for **people tracking, generic industrial DT architecture, and alarm-management principles**, but weaker for **forklift-specific MTMC**, **factory-floor homography head-to-head comparisons**, **multi-site camera-fleet scaling metrics**, and **longitudinal safety outcomes** [4] [22] [27] [43]. Some deployment conclusions therefore require cautious inference from adjacent construction, mining, healthcare-alarm, and industrial monitoring literature [28] [30] [38].

Suggested Further Research

- Run a **factory trial** that jointly reports localization error, IDF1/HOTA, FAR, miss rate, alarm delay, and operator response in one deployment [1] [7] [23].
- Build a **forklift-focused MTMC benchmark** for overlapping and non-overlapping industrial cameras [22] [59].
- Quantify **calibration drift intervals and labor burden** in industrial CCTV fleets [11] [57] [60].
- Benchmark **many-camera alert freshness and degraded-mode behavior** under overload and network loss [40] [41] [43].
- Publish **factory-specific governance implementations** covering retention, worker notice, access control, and audit logging [45] [46] [47].

Full Report

Foundational Context

FactoryFlow combines four technical layers: **detection**, **tracking**, **camera-to-world projection**, and **safety decision logic**. In deployment, failures often arise not from one layer alone but from error propagation across the stack [32] [61] [62]. A missed or shifted detection can fragment a track; a fragmented track can break cross-camera identity; a broken identity or misprojected footpoint can trigger a wrong proximity alert on the dashboard [1] [15] [26] [63].

Two distinctions matter:

1. **Safety zoning vs precise localization.** The literature more strongly supports sub-meter localization for conservative hazard-zone classification than for precise digital-twin coordinates suitable for trusted trajectory analytics or simulation inputs [23] [28] [29].
2. **Perception validation vs safety validation.** Good detector/tracker scores do not establish that a warning system has acceptable false alarms, delay, human response, or incident reduction [1] [4] [7].

Confidence level: High - these framing distinctions are directly supported across the localization and alarm-management evidence [7] [29].

1. Ranked deployment gaps: proven blockers first

1.1 Highest-confidence blockers

1) Safety validation gap

The most important blocker is not raw model accuracy but the absence of a full operational safety case. The closest retrieved vision-based proximity system reports strong partial metrics—93.75% average proximity accuracy, 97.5% precision for static hazards, dynamic-hazard hit rates of 85.7%-89.4%, miss rates of 10.5%-14.2%, and four false alarms—but this still does not amount to a complete factory deployment validation package [1]. The broader corpus explicitly states that no retrieved paper reports precision, recall, false-alarm rate, miss rate, and operator response time together, and no factory/warehouse field trial shows longitudinal reductions in injuries or incidents from camera-based analytics [2] [3] [4] [64].

Fix: adopt industrial alarm metrics such as FAR, MAR, and average alarm delay, then test staged/persistence-based warnings in field trials with supervisor-verified labels and response logging [1] [6] [7] [34].

2) Calibration drift and maintenance burden

Drift is directly evidenced. Real-time homography updates have been needed after slight camera view shifts [8]. Temperature changes alter geometry and require correction [9]. Vibration, impacts, relocation, and hardware changes can invalidate extrinsics over time [10] [11]. Periodic recalibration is repeatedly recommended, but large-fleet intervals and labor costs are poorly quantified [57] [60].

Fix: online drift monitoring, self-calibration from natural features or line cues, plus explicit calibration-health checks [11] [13] [14].

3) Cross-camera identity persistence

The evidence consistently shows that geometry plus appearance improves MTMC, but identity persistence remains brittle under occlusion, similar uniforms, viewpoint change, lighting variation, and fragmented trajectories [15] [16] [21]. Online tracking is particularly vulnerable because it cannot use future frames [17]. Geometry-aware real-time tracking on campus/factory data reached 61.8% IDF1 and 54.3% MOTA across four cameras, which is useful but far from “solved” [15].

Fix: use homography-based matching in overlapping views, Re-ID plus spatio-temporal gating for blind spots, global ID pools, and selective trajectory re-evaluation [18] [19] [20] [65].

4) Localization accuracy for floor-plan mapping

Projected localization can be good enough for zones: one indoor study reported 13.2 cm mean Euclidean error and judged it acceptable for hazard-zone detection [23]. Another reported average inter-person distance error around 0.1 m and max pedestrian error within 0.25 m [24]. But another found errors up to 1 m in farther image regions, and error rose with worker-camera distance and occlusion [25] [26]. The retrieved set does not provide a factory-specific breakdown by camera height, tilt, occlusion, and forklift motion [27] [66].

Fix: use conservative hazard buffers, validate against surveyed floor points, and add fiducials or anchor references in critical areas [28] [29] [67].

5) False-alarm management

The alarm literature is unequivocal that nuisance alerts harm compliance. In healthcare and other safety-critical domains, nonactionable alarms have consumed most alerts and delayed responses; ignored false alarms are directly documented [30] [31]. Industrial alarm-management papers treat false alarms, missed alarms, deadbands, and delays as central design trade-offs [7] [32] [68].

Fix: persistence windows, yellow/red intervention zones, deadbands, and adaptive thresholds instead of single-frame triggering [33] [34] [35].

Confidence level: High - these are repeatedly observed failure modes with direct deployment implications [1] [10] [15] [30].

1.2 Proven gaps versus assumptions

Proven in the literature

- Calibration drift from environmental and mechanical change [8] [9] [10].
- Localization degradation with distance, angle, and occlusion [25] [26].
- Fragile cross-camera identity for people [15] [21].
- False-alarm harms and need for alarm governance [7] [30] [31].
- Thin evidence for many-camera latency/reliability benchmarking [39] [43].
- Privacy and governance controls are required but incompletely operationalized in factory CCTV deployments [46] [47].

Assumptions or weakly evidenced

- Forklift cross-camera ID persistence under real factory CCTV layouts [22] [59] [69].
- Exact recalibration frequency for factory fleets [57] [60].
- Sustained dashboard freshness at large camera counts [43] [54].

- Actual injury reduction attributable to camera-based proximity analytics [4] [5].

Confidence level: High - the evidence gap boundaries are explicit in the retrieved corpus [4] [22] [27] [43].

2. Localization and floor-plan mapping limits

Homography and related floor-plane projections are attractive because they are simple and cheap, but their deployment suitability is conditional [70] [71]. Reported errors span a wide range: 13.2 cm mean error in a controlled indoor hazard-zone study [23], around 0.1 m average inter-person distance error in social-distancing monitoring [24], and up to 1 m error in far image regions in another indoor study [25]. Error increases with distance and occlusion, and camera geometry materially affects results [25] [26] [72].

The literature does support a practical deployment interpretation: **for warning zones, conservative buffers can absorb moderate position error** [28] [29]. Field warning studies have used threshold logic such as 2 m and 2 s rather than relying on exact coordinates [73]. Taken together, these findings suggest that FactoryFlow should initially claim **hazard-zone occupancy confidence**, not high-precision digital twin localization [28] [29].

What is missing is equally important: no retrieved study gives a strong 2020-2025 factory-floor head-to-head comparison of planar homography versus fiducials, bundle adjustment, or camera-LiDAR alignment for YOLO-based worker/forklift projection [27] [74]. Marker-based alternatives do provide stable reference structure, but the corpus does not show decisive deployment-side comparisons on repeatability or setup burden [67] [75].

Implication: localization is not the main blocker if the use case is conservative safety zoning, but it becomes a blocker if the product promise includes precise trajectory analytics, simulation fidelity, or reliable near-miss geometry across varied camera placements [23] [25] [27].

Confidence level: High - direct error evidence is strong, but factory-specific decomposition is missing [23] [25] [27].

3. Calibration drift and maintenance burden

This is one of the clearest operational blockers. Multiple sources identify causes of calibration decay: view shifts, vibration, impact, thermal drift, relocation, environmental instability, and scene change [8] [9] [10] [12]. Even when cameras remain nominally fixed, homography accuracy can degrade under lighting, perspective, and blur changes [76].

The strongest fixes are not exotic model changes but **maintenance automation**. Online camera-LiDAR extrinsic monitoring can track drift with about 0.03° yaw accuracy and >98% monitoring accuracy [13]. Built-environment line cues can support reliable online recalibration while handling uncertainty and frame selection [14]. Framework papers explicitly recommend automated self-calibration, natural scene references, and periodic calibration-health checks [11].

What remains uncertain is lifecycle burden at fleet scale. The corpus shows that recalibration is commonly redone after camera repositioning or setting changes, and periodic schedules vary by application from every two weeks to monthly, but there is no factory CCTV consensus interval [77] [78] [79].

Practical fix ranking

1. Detect drift continuously and score each camera's mapping trustworthiness [11] [13].
2. Recalibrate automatically from scene lines/features where possible [11] [14].
3. Use surveyed floor checkpoints in critical aisles/hazard zones [67] [80].
4. Downgrade cameras with failed calibration health to advisory-only status until refreshed [11] [41].

Confidence level: High - drift is directly proven; exact maintenance cost is not [10] [57].

4. Identity persistence across cameras and occlusions

Per-camera ByteTrack-style tracking is not enough for a factory digital twin that claims near-miss chains, heatmaps, or simulation-ready trajectories across blind spots. Single-camera MOT has persistent failure modes from missing detections, occlusion, and weak Re-ID, even with strong baselines such as SORT and Deep SORT derivatives [61] [63] [81] [82].

For multi-camera tracking, the evidence supports a **hybrid geometry + appearance** approach. The MTA baseline combines appearance distance, camera homographies, overlap-time constraints, and linear prediction, yielding a +13% IDF1 gain [20]. SpotOn uses homography-based floor-plan matching for overlapping cameras and Re-ID only for blind spots, explicitly targeting campus/factory settings [15]. Global ID logic can greatly reduce switches in other domains, with one ground-plane/global-pool system cutting raw ID switches by 87.6% [65]. Risk-aware selective re-evaluation and accumulated appearance plus trajectory validation also improve identity consistency with low overhead [18] [19].

However, failures remain common under similar clothing, illumination shifts, occlusion, viewpoint changes, low resolution, and fragmented trajectories [16] [21] [83] [84]. The forklift case is the major evidence gap: the retrieved set does not include direct 2019-2025 forklift-focused homography-plus-Re-ID MTMC evaluations [22] [59] [69].

Inferred: forklift identity may be more recoverable than worker identity if deployments add weak asset cues such as fixed vehicle IDs or telemetry hooks, because geometry-only and appearance-only association are both documented as brittle [85] [86] [87]. **Confidence level: Low** - this is a reasoned inference, not a direct finding.

Confidence level: High for people, Low for forklifts - human MTMC blockers are well evidenced; forklift persistence is an explicit evidence gap [15] [22].

5. Safety validation, false alarms, and near-miss claims

The corpus draws a sharp line between detecting a hazard and proving a safety outcome. Real-world proximity systems do report useful partial metrics and field testing [1] [2], but the retrieved literature does not provide a single complete deployment study with precision, recall, false-alarm rate, miss rate, alert delay, operator response, and longitudinal reduction in incidents [2] [4].

This matters because near-miss systems only improve safety when organizations respond effectively. A prospective cohort found that adequate company response to near-miss reporting was associated with fewer later accidents [88]. By contrast, quota-driven reporting can increase counts without improving outcomes [89]. So a dashboard full of proximity events is not evidence of safety benefit unless the alarm process is actionable and acted on [90] [91].

False alarms are especially dangerous for deployment credibility. Across domains, nuisance alerts cause desensitization, ignored warnings, and delayed response [30] [31] [92]. Persistence-based confirmation, multi-frame verification, and staged zones are repeatedly proposed or evaluated as ways to lower nuisance burden while maintaining detection [1] [33] [34] [35].

Practical validation package for FactoryFlow

- Event-level **FAR, MAR, average alarm delay** [7].
- **Alert freshness** from first visible hazard to dashboard/alarm [1] [42].
- **Persistence/staging comparisons** against single-frame alerts [1] [34].
- **Supervisor-verified ground truth** on a sampled set [3].
- **Management response audit** linking alerts to actions taken [88].

Confidence level: High - the literature is consistent that alarm design and operational follow-through matter as much as perception quality [7] [30] [88].

6. Reliability, latency, and degraded-mode operation

The architecture literature favors edge/fog/hybrid DTs because they reduce latency, bandwidth use, and dependence on unstable connectivity [37] [38] [93]. That general direction is well supported. What is less supported is the specific claim that a many-camera CCTV safety twin will maintain acceptable alert freshness under overload, network loss, or component failures [39] [43].

The most relevant reliability evidence is indirect but important. Testudo uses redundant local execution as a fail-safe under wireless-link uncertainty and reports little-to-no prediction-score degradation while adapting to network status [41]. An empirical edge fault-injection study identifies memory overload, cache thrashing, context switching, and page faults as major latency disruptors [40]. A traffic-safety edge system kept average latency below 58 ms, but maximum spikes reached 800 ms under load [42].

Taken together, these findings suggest that **average latency is an insufficient deployment metric**; tail latency and overload behavior matter more for safety alerts [40] [42].

Recommended runtime controls

- Edge-first inference for safety-critical camera streams [38] [94].
- Local failover/degraded mode if connectivity drops [41].
- Buffers and explicit backpressure management [40].
- Service-level objectives for p95/p99 alert delay and dropped-frame tolerance [42] [43].

Confidence level: Medium - architecture direction is well supported, but factory CCTV-specific benchmarks are scarce [37] [43].

7. Privacy, governance, and workplace acceptability

Governance is not an optional add-on. The privacy literature strongly converges on transparency, data minimization, architectural decoupling, spatial aggregation, role-based access, audit logging, and limited retention [44] [45] [46]. Worker notice and accessible privacy policies are repeatedly emphasized, and consent alone may be insufficient because of power imbalance in employment settings [48] [95] [96].

But implementation evidence is thin. The retrieved set explicitly lacks a complete industrial CCTV case that covers worker consent, retention schedules, access-control rules, and auditing procedures in one deployed system [47] [97] [98]. Even feasibility can fail at the infrastructure level: one food-manufacturing CCTV study excluded two facilities because coverage was insufficient [99].

For FactoryFlow, the minimum viable governance design is therefore clear:

- On-device or edge inference where possible [100] [101].

- Role-based access to identifiable views/data [45].
- Limited retention and explicit purpose control [46].
- Worker-facing transparency and prior notice [48] [95].
- Audited access and policy enforcement across sites [102] [103].

Confidence level: High - governance requirements are consistent across sources, though factory implementation detail is sparse [44] [47].

8. Scale, interoperability, and lifecycle operations

The evidence suggests that scale-out is blocked less by any single detector and more by systems integration. Reviews emphasize standardized interfaces, semantic interoperability, edge/cloud data fusion, and orchestration as prerequisites for smart-factory DT scaling [50] [104]. At the same time, they repeatedly report hardware heterogeneity, network congestion, protocol incompatibility, data integration issues, and high implementation cost/time as practical barriers [51] [105] [106].

Standards help, but the evidence for measured reduction in integration burden is modest. ISO 23247, AAS, and OPC UA are presented as enablers, yet adoption is incomplete and semantic gaps remain [52] [107] [108]. For vision specifically, machine-vision interface standards reduce vendor lock-in and custom integration effort at the camera/computer layer [109] [110] [111]. Direct evidence connecting camera analytics with MES/SCADA/maintenance end to end remains sparse [109] [112].

Lifecycle burden is framed more clearly around **model drift, retraining, and data throughput** than around recalibration alone. One study recommends tracking model-update frequency, retraining dataset volume, and drift metrics across time [53]. Another notes that high-frequency sensors can exceed practical throughput budgets, forcing trade-offs such as downsampling or edge feature extraction [113].

Microservice architectures are generally favored over monoliths for maintainability, resilience, and fault isolation, but no retrieved paper directly compares them for a large-scale factory-monitoring dashboard with many camera streams and analytics modules [55] [56] [104].

Deployment implication: FactoryFlow's scale risk is primarily operational: orchestration, interface contracts, retraining/drift workflows, and throughput governance must be designed from the start [51] [53] [55].

Confidence level: High for the existence of scale barriers; Medium for specific architecture superiority in camera-heavy twins [51] [104].

Conclusion

The literature supports a clear deployment posture for FactoryFlow:

- **Most proven blockers:** incomplete safety validation, calibration drift, fragile cross-camera identity, false alarms, and systems-scale integration burdens [4] [10] [15] [31] [51].
- **Most defensible near-term product claim:** a **conservative safety-zoning and alerting system**, not a precise factory localization or simulation-truth system [28] [29].
- **Most evidence-backed fixes:** persistence/staged alerts, calibration-health monitoring with online correction, geometry-plus-ReID cross-camera association, edge-first degraded-mode runtime design, and privacy-by-design governance [14] [20] [34] [41] [44].
- **Main assumptions still requiring direct proof:** forklift MTMC, exact recalibration intervals, large-camera-fleet dashboard freshness, and actual injury reduction in factories [4] [22] [43] [57].

If FactoryFlow is deployed without solving those first-order gaps, the likely failure mode is not that detection never works, but that the system becomes **operationally untrusted**: coordinates drift, IDs break across cameras, alarms become noisy, and safety benefit remains unproven [4] [8] [16] [30].

Overall confidence level: High - the corpus consistently supports the priority ordering of blockers, though exact quantitative thresholds for factory-scale deployment remain underreported [4] [10] [15] [43] [51].

References

1. Jeelani, I., Asadi, K., Ramshankar, H., (...), Albert, A. (2021) Real-time vision-based worker localization & hazard detection for construction. *Automation in Construction*. https://doi.org/10.1016/j.autcon.2020.103448
2. Chan, Kelsey, Louis, Joseph, Albert, Alex (2020) Incorporating Worker Awareness in the Generation of Hazard Proximity Warnings. *Sensors (Basel, Switzerland)*. https://doi.org/10.3390/s20030806
3. Jiang, W., Liu, Y., Chen, K., (...), Ding, L. (2024) Early-warning of unsafe hoisting operations: An integration of digital twin and knowledge graph. *Developments in the Built Environment*. https://doi.org/10.1016/j.dibe.2024.100490
4. Gnoni, M.G., Elia, V., Tornese, F., (...), Pellicci, M. (2024) A web-based platform to support near miss management systems in industrial companies: the CONDIVIDO tool. *IEEE International Conference on Industrial Engineering and Engineering Management*. https://doi.org/10.1109/IEEM62345.2024.10857025
5. WoźNiak, Z., Hola, B. (2024) Analysing Near-Miss Incidents in Construction: A Systematic Literature Review. *Applied Sciences (Switzerland)*. https://doi.org/10.3390/app14167260
6. Weng, X., Xu, X., Wang, X., (...), Zhou, Z. (2025) A hybrid stochastic-dynamic framework for reliable alarm generation: Integrating GMM-based probability patterns with temporal evidence fusion. *Chemical Engineering Research and Design*. https://doi.org/10.1016/j.cherd.2025.08.043

7. Xu, J., Wang, J., Izadi, I., Chen, T. (2012) Performance assessment and design for univariate alarm systems based on FAR, MAR, and AAD. *IEEE Transactions on Automation Science and Engineering*. https://doi.org/10.1109/TASE.2011.2176490
8. Yang, B., Gao, B., Han, Y., Zhang, B. (2023) Semantic segmentation-based framework for concrete pouring progress monitoring by using multiple surveillance cameras. *Developments in the Built Environment*. https://doi.org/10.1016/j.dibe.2023.100283
9. Elias, Melanie, Eltner, Anette, Liebold, Frank, Maas, Hans-Gerd (2020) Assessing the Influence of Temperature Changes on the Geometric Stability of Smartphone- and Raspberry Pi Cameras. *Sensors (Basel, Switzerland)*. https://doi.org/10.3390/s20030643
10. He, Z., Tan, J., Lin, Z., (...), Li, E. (2025) Online extrinsic parameters calibration of on-board stereo cameras based on certifiable optimization. *Measurement: Journal of the International Measurement Confederation*. https://doi.org/10.1016/j.measurement.2024.115911
11. Honarparvar, Sepehr, Honarparvar, Yasaman, Ashena, Zahra, (...), Saeedi, Sara (2025) GICEDCam: A Geospatial Internet of Things Framework for Complex Event Detection in Camera Streams. *Sensors (Basel, Switzerland)*. https://doi.org/10.3390/s25175331
12. Chen, Zhichao, Chen, Lifang, Gong, Jianjun, (...), Yang, Jie (2027) A structured review of railway obstacle detection as an engineering informatics problem: Sensing, representation, decision support, and traceability. *Advanced Engineering Informatics*. https://doi.org/10.1016/j.aei.2026.105337
13. Moravec, J., Šára, R. (2024) Online Camera–LiDAR Calibration Monitoring and Rotational Drift Tracking. *IEEE Transactions on Robotics*. https://doi.org/10.1109/TRO.2023.3347130
14. Qian, Y., Elder, J.H. (2022) A Reliable Online Method for Joint Estimation of Focal Length and Camera Rotation. *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*. https://doi.org/10.1007/978-3-031-19769-7_15
15. Setdhavanich, K., Jeraja, Y., Erjongmanee, S., (...), Klumpol, C. (2026) SpotOn: A Geometry-Aware Multi-Camera Person Tracking System for Real-Time Situational Awareness. *Proceedings - 23rd International Joint Conference on Computer Science and Software Engineering, JCSSE 2026*. https://doi.org/10.1109/JCSSE68839.2026.11597011
16. Cunico, F., Cristani, M. (2025) Multi-camera Industrial Open-Set Person Re-identification and Tracking. *Lecture Notes in Computer Science*. https://doi.org/10.1007/978-3-031-91575-8_8
17. Liu, Y., Li, X., Bai, T., (...), Wang, F.-Y. (2021) Multi-object tracking with hard-soft attention network and group-based cost minimization. *Neurocomputing*. https://doi.org/10.1016/j.neucom.2021.02.084
18. Huang, Y., Li, C., Hu, J., Zhang, X. (2026) Risk-Aware Identity Management for Online Multi-Camera Vehicle Tracking. *2026 2nd International Conference on Unmanned Systems and Technology, UST 2026*. https://doi.org/10.1109/UST70584.2026.11663771

19. Tran, L.Q., Duc Vi, H. (2024) Efficient Online Multi-Camera Tracking with Memory-Efficient Accumulated Appearance Features and Trajectory Validation. *IEEE Computer Society Conference on Computer Vision and Pattern Recognition Workshops*. https://doi.org/10.1109/CVPRW63382.2024.00717

20. Kohl, P., Specker, A., Schumann, A., Beyerer, J. (2020) The MTA dataset for multi target multi camera pedestrian tracking by weighted distance aggregation. *IEEE Computer Society Conference on Computer Vision and Pattern Recognition Workshops*. https://doi.org/10.1109/CVPRW50498.2020.00529

21. Kegeleirs, Miquel, Gharbi, Ilyes, Kaplanis, Marios, (...), Birattari, Mauro (2025) Assessing the impact of feature communication in swarm perception for people re-identification. *Frontiers in Robotics and AI*. https://doi.org/10.3389/frobt.2025.1671952

22. Zhang, X., Xie, X., Wen, L., Lai, J. (2024) Person group detection with global trajectory extraction in a disjoint camera network. *Neurocomputing*. https://doi.org/10.1016/j.neucom.2024.127281

23. Kulinan, A.S., Park, M., Aung, P.P.W., (...), Park, S. (2024) Advancing construction site workforce safety monitoring through BIM and computer vision integration. *Automation in Construction*. https://doi.org/10.1016/j.autcon.2023.105227

24. Sathyamoorthy, Adarsh Jagan, Patel, Utsav, Paul, Moumita, (...), Manocha, Dinesh (2021) COVID surveillance robot: Monitoring social distancing constraints in indoor scenarios. *PLoS ONE*. https://doi.org/10.1371/journal.pone.0259713

25. Rahimi, M.M., Khoshelham, K., Stevenson, M., Winter, S. (2021) Pose-aware monocular localization of occluded pedestrians in 3D scene space. *ISPRS Open Journal of Photogrammetry and Remote Sensing*. https://doi.org/10.1016/j.ophoto.2021.100006

26. Goh, Y.M., Tian, J., Chian, E.Y.T. (2022) Management of safe distancing on construction sites during COVID-19: A smart real-time monitoring system. *Computers and Industrial Engineering*. https://doi.org/10.1016/j.cie.2021.107847

27. Assadzadeh, Amin, Arashpour, Mehrdad, Bab-Hadiashar, Alireza, (...), Li, Heng (2021) Automatic far-field camera calibration for construction scene analysis. *Computer-Aided Civil and Infrastructure Engineering*. https://doi.org/10.1111/mice.12660

28. Edrei, T., Isaac, S. (2017) Construction site safety control with medium-accuracy location data. *Journal of Civil Engineering and Management*. https://doi.org/10.3846/13923730.2016.1144644

29. Park, J., Cho, Y.K., Khodabandelu, A. (2018) Sensor-based safety performance assessment of individual construction workers. *Sensors (Switzerland)*. https://doi.org/10.3390/s18113897

30. Stiglich, Y.F., Brener Dik, P.H.B., Segura, M.S., Mariani, G.L. (2023) The Alarm Fatigue Challenge in the Neonatal Intensive Care Unit: A before and after Study. *American Journal of Perinatology*. https://doi.org/10.1055/a-2113-8364

31. Abdul Rahman, A.R., Aswad, A., Muhammad, M., (...), Yahya, Y. (2012) The response of theatre personnel to the pulse oximeter alarm. *Southern African Journal of Anaesthesia and Analgesia*. https://doi.org/10.1080/22201173.2012.10872834

32. Izadi, I., Shah, S.L., Shook, D.S., (...), Chen, T. (2009) A framework for optimal design of alarm systems. *IFAC Proceedings Volumes (IFAC-PapersOnline)*. https://doi.org/10.3182/20090630-4-es-2003.00108

33. Nogueira, Welker Facchini, Michalski, Miguel Angelo de Carvalho, Melani, Arthur Henrique de Andrade, (...), de Souza, Gilberto Francisco Martha (2026) Anomaly Detection in Wind Turbines: Persistence-Based Alarm Confirmation for False-Alarm Mitigation and Detection-Latency Trade-Offs. *Sensors (Basel, Switzerland)*. https://doi.org/10.3390/s26123896

34. https://www.sciencedirect.com/science/article/pii/S1863171125009902

35. Sanchez, O.T., Borges, G., Raposo, D., (...), Silva, J.S. (2025) Enhancing the performance of Industrial IoT LoRaWAN-enabled federated learning frameworks: A case study. *Internet of Things (The Netherlands)*. https://doi.org/10.1016/j.iot.2025.101632

36. Kondaveeti, S.R., Izadi, I., Shah, S.L., (...), Chen, T. (2012) Graphical tools for routine assessment of industrial alarm systems. *Computers and Chemical Engineering*. https://doi.org/10.1016/j.compchemeng.2012.06.042

37. Abdullahi, Ibrahim, Longo, Stefano, Samie, Mohammad (2024) Towards a Distributed Digital Twin Framework for Predictive Maintenance in Industrial Internet of Things (IIoT). *Sensors (Basel, Switzerland)*. https://doi.org/10.3390/s24082663

38. Shafik, Wasswa (2025) Digital twins tools and technologies. *Digital Twins for Smart Cities and Villages*. https://doi.org/10.1016/B978-0-443-28884-5.00003-8

39. Deng, R., Zhu, S., Lu, T., (...), Usmani, A. (2026) Vision-driven dynamic tunnel digital twin for real-time situation and emergency awareness. *Underground Space (new)*. https://doi.org/10.1016/j.undsp.2026.05.004

40. Pourreza, M., Narasimhan, P. (2023) An Empirical Study of Resource-Stressing Faults in Edge-Computing Applications. *EdgeSys 2023 - Proceedings of the 6th International Workshop on Edge Systems, Analytics and Networking, Part of EuroSys 2023*. https://doi.org/10.1145/3578354.3592873

41. Odema, M., Chen, L., Levorato, M., Al-Faruque, M.A.A. (2023) Testudo: Collaborative Intelligence for Latency-Critical Autonomous Systems. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*. https://doi.org/10.1109/TCAD.2022.3211480

42. Chebaane, A., Scholze, D., Rezgui, Y., (...), Liu, C. (2025) Fine-grained latency analysis of real-world 5G-enabled multi-tier edge computing for school zone traffic safety. *Discover Internet of Things*. https://doi.org/10.1007/s43926-025-00218-1

43. Elias, C., Hutchins, I., Issa, R.R.A. (2026) OPERATIONAL DIGITAL TWINS IN THE BUILT ENVIRONMENT: A SYSTEMATIC REVIEW OF SEMANTIC INFRASTRUCTURE AND DEPLOYMENT PRACTICE. *Journal of Information Technology in Construction*. https://doi.org/10.36680/j.itcon.2026.043
44. Renken, S., Combe, A.L. (2026) Design Principles for Privacy Preserving IoT Workplace Monitoring. *Pacific Asia Conference on Information Systems*. https://www.scopus.com/pages/publications/105046130116
45. Plantes, Peter J., Fragala, Maren S., Clarke, Charles, (...), Goldberg, Steven E. (2021) Model for Mitigation of Workplace Transmission of COVID-19 Through Population-Based Testing and Surveillance. *Population Health Management*. https://doi.org/10.1089/pop.2020.0322
46. Das, N., Murthy, G.S.N., Akhila, K., (...), Nagender (2025) FD-Net: Fire detection from CC videos with frame inter correlated pattern analysis and Bi-directional SVM pyramid networks. *Systems and Soft Computing*. https://doi.org/10.1016/j.sasc.2025.200393
47. Piza, E.L. (2021) The History, Policy Implications, and Knowledge Gaps of the CCTV Literature: Insights for the Development of Body-Worn Video Camera Research. *International Criminal Justice Review*. https://doi.org/10.1177/1057567718759583
48. Wolff, M.S., Ravid, D.M., Behrend, T.S. (2025) Guidelines for the use of electronic performance monitoring. *Organizational Dynamics*. https://doi.org/10.1016/j.orgdyn.2023.101026
49. https://www.sciencedirect.com/science/article/pii/S1932226722000162
50. Leong, W.Y. (2025) Digital Twin-Enhanced Communication Architectures for Malaysia's Smart Manufacturing Sector. *Proceedings - 2025 RIVF International Conference on Computing and Communication Technologies, RIVF 2025*. https://doi.org/10.1109/RIVF68649.2025.11365091
51. Judijanto, L., Nurhayati, S. (2025) Robotic Digital Twins in Smart Manufacturing: Resolving Real-Time Synchronisation Gaps. *International Conference on Computer and Applications, ICCA 2025 - Proceedings*. https://doi.org/10.1109/ICCA66035.2025.11430951
52. Huebner, J.E., Ihlenfeldt, S. (2026) SEMANTIC INTEROPERABILITY IN INDUSTRY 4.0: A SYSTEMATIC MAPPING STUDY ON INTEGRATING KNOWLEDGE GRAPHS WITH AAS AND OPC UA. *Journal of Machine Engineering*. https://doi.org/10.36897/jme/219827
53. Yari Kalashgrani, M., Sattarpanah Karganroudi, S., Barangizi, H., (...), Lai, C.W. (2026) Artificial intelligence-assisted three-dimensional printing of lithium-ion batteries: Improving quality control in energy storage systems. *Journal of Energy Storage*. https://doi.org/10.1016/j.est.2026.123518
54. de Larmelina, S., da Silva, A.L., Risso, L.A. (2026) Conceptualizing the Digital Twin: An Analysis of Definitions, Applications, Technologies, Benefits and Challenges. *Journal of Advanced Manufacturing Systems*. https://doi.org/10.1142/S0219686726500125
55. Wang, Y.-T., Ma, S.-P., Lai, Y.-J., Liang, Y.-C. (2022) Analyzing and Monitoring Kubernetes Microservices based on Distributed Tracing and Service Mesh. *Proceedings - Asia-Pacific Software Engineering Conference, APSEC*. https://doi.org/10.1109/APSEC57359.2022.00066

56. Somla, Srinivas, S. (2026) Large-Scale Intelligence Microservices For Robust Anomaly Detection In Real-Time Streaming Data Systems Using Deep Learning. *International Journal of Artificial Intelligence and Machine Learning*. https://doi.org/10.51483/ijaiml.6.2s.2026.300-309
57. Ruchay, A., Kolpakov, V., Ryazanov, V., (...), Pezzuolo, A. (2026) Real-Time Automated Body Condition Scoring of Dairy Cows in a Commercial Rotary Milking Parlor Using RGB-D Two-Stream Deep Learning. *Computers and Electronics in Agriculture*. https://doi.org/10.1016/j.compag.2026.112252
58. Mirosławski, T. (2025) WORKPLACE MONITORING IN EUROPE: A REVIEW OF REGULATORY PERSPECTIVES AND DEVELOPMENTS. *Polish Review of International and European Law*. https://doi.org/10.21697/2025.14.2.04
59. https://www.sciencedirect.com/science/article/pii/S1573749723005441
60. Cofta, P., Marciniak, B. (2024) The effectiveness of reference nodes in reputation-based ad-hoc sensor networks. *AEU - International Journal of Electronics and Communications*. https://doi.org/10.1016/j.aeue.2024.155244
61. Bewley, A., Ge, Z., Ott, L., (...), Upcroft, B. (2016) Simple online and realtime tracking. *Proceedings - International Conference on Image Processing, ICIP*. https://doi.org/10.1109/ICIP.2016.7533003
62. Vo, T.D.K., Jeon, Y., Pham, T.-D., (...), Park, S. (2026) Camera-relocalization-driven monocular 3D safety monitoring system for crane operations. *Automation in Construction*. https://doi.org/10.1016/j.autcon.2026.107057
63. Bergmann, P., Meinhardt, T., Leal-Taixé, L. (2019) Tracking without bells and whistles. *Proceedings of the IEEE International Conference on Computer Vision*. https://doi.org/10.1109/ICCV.2019.00103
64. Nnaji, C., Gambatese, J., Lee, H.W., Zhang, F. (2020) Improving construction work zone safety using technology: A systematic review of applicable technologies. *Journal of Traffic and Transportation Engineering (English Edition)*. https://doi.org/10.1016/j.jtte.2019.11.001
65. Zhao, B., Xiao, B., Liu, C., (...), Yu, Z. (2026) Bovine-LTID: A vision-based spatio-temporal identity system for multi-camera cattle tracking and long-term behavior monitoring. *Computers and Electronics in Agriculture*. https://doi.org/10.1016/j.compag.2026.112051
66. https://www.sciencedirect.com/science/article/pii/S143330582500379X
67. https://www.sciencedirect.com/science/article/pii/S1433301520006154
68. Adnan, N.A., Izadi, I., Chen, T. (2011) On expected detection delays for alarm systems with deadbands and delay-timers. *Journal of Process Control*. https://doi.org/10.1016/j.jprocont.2011.06.019
69. Wang, X., Li, D., Wang, J., (...), Song, B. (2025) Continuous multi-target tracking across disjoint camera views for field transport productivity analysis. *Automation in Construction*. https://doi.org/10.1016/j.autcon.2025.105984

70. Lv, Hanming, Teng, Fuhong, Wu, Yutao (2024) Measurement of ring spinning spun yarn spindle position based on binocular vision. *Textile Research Journal*. https://doi.org/10.1177/00405175241235399

71. Gao, B., Yang, B., Xiao, H., Zhou, Y. (2025) Production-based progress monitoring of rebar tying using few-shot learning and kernel density. *Alexandria Engineering Journal*. https://doi.org/10.1016/j.aej.2024.12.098

72. Iusupova, A., Titov, V. (2022) Empirical Analysis of Visual Localization Accuracy. *Proceedings - 2022 International Russian Automation Conference, RusAutoCon 2022*. https://doi.org/10.1109/RusAutoCon54946.2022.9896298

73. Luo, X., Li, H., Huang, T., Rose, T. (2016) A field experiment of workers' responses to proximity warnings of static safety hazards on construction sites. *Safety Science*. https://doi.org/10.1016/j.ssci.2015.12.026

74. https://www.sciencedirect.com/science/article/pii/S1433301520006154

75. Park, S., Hong, C., Hwang, I., Lee, J. (2023) Comparison of Single-Camera-Based Depth Estimation Technology for Digital Twin Model Synchronization of Underground Utility Tunnels. *Applied Sciences (Switzerland)*. https://doi.org/10.3390/app13042106

76. Ahmed, M.W., Adnan, M., Ahmed, M., (...), Ectors, W. (2025) A lightweight georeferencing workflow for dynamic UAV footage using feature-matching and minimal drone metadata. *Remote Sensing Applications: Society and Environment*. https://doi.org/10.1016/j.rsase.2025.101801

77. Janisch, Judith, Mitoyen, Clementine, Perinot, Elisa, (...), Quigley, Cliodhna (2021) Video Recording and Analysis of Avian Movements and Behavior: Insights from Courtship Case Studies. *Integrative and Comparative Biology*. https://doi.org/10.1093/icb/icab095

78. Campbell, Tanner, Battle, Adam, Gray, Dan, (...), Furfaro, Roberto (2024) Stingray Sensor System for Persistent Survey of the GEO Belt. *Sensors (Basel, Switzerland)*. https://doi.org/10.3390/s24082596

79. Shi, Baiqian, Catsamas, Stephen, Kolotelo, Peter, (...), McCarthy, David T. (2021) A Low-Cost Water Depth and Electrical Conductivity Sensor for Detecting Inputs into Urban Stormwater Networks. *Sensors (Basel, Switzerland)*. https://doi.org/10.3390/s21093056

80. Pak, A., Reichel, S., Burke, J. (2022) Machine-Learning-Inspired Workflow for Camera Calibration. *Sensors*. https://doi.org/10.3390/s22186804

81. Wojke, N., Bewley, A., Paulus, D. (2017) Simple online and realtime tracking with a deep association metric. *Proceedings - International Conference on Image Processing, ICIP*. https://doi.org/10.1109/ICIP.2017.8296962

82. Pang, B., Li, Y., Zhang, Y., (...), Lu, C. (2020) Tubetk: Adopting tubes to track multi-object in a one-step training model. *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition*. https://doi.org/10.1109/CVPR42600.2020.00634

83. Cai, J., He, Z., Liu, Z., Cao, Y. (2026) DiPerceiveNet: A bidirectional cross-scale perception network for vehicle re-identification. *Pattern Recognition*. https://doi.org/10.1016/j.patcog.2026.113476

84. Gellida-Coutino, C., Rios-Cabrera, R., Maldonado-Ramirez, A., Sanchez-Orta, A. (2025) Real-Time Multi-Camera Tracking for Vehicles in Congested, Low-Velocity Environments: A Case Study on Drive-Thru Scenarios. *Electronics (Switzerland)*. https://doi.org/10.3390/electronics14132671

85. El-Alami, A., Nadir, Y., Mansouri, K. (2025) A Distributed Approach of Multi-camera Vehicle Tracking System Based on Geometric Association. *Lecture Notes in Networks and Systems*. https://doi.org/10.1007/978-3-031-95326-2_28

86. Alenezi, A., Alabaiadly, Y. (2026) Digital Twins and Edge AI for Small/Off-Grid Water Systems: A Review. *IEEE Access*. https://doi.org/10.1109/ACCESS.2026.3669429

87. Sharma, A., Mallala, N. (2026) Safety-Constrained Multimodal Digital Twins for Zero-Defect Industrial Workcells. *2026 6th International Conference on Intelligent Technologies, CONIT 2026*. https://doi.org/10.1109/CONIT69683.2026.11621740

88. Inagaki, M., Nagata, T., Odagami, K., (...), Mori, K. (2024) Relationship between a company's adequate response to near-misses and occupational accidents: a 1-year prospective cohort study. *Journal of Occupational Health*. https://doi.org/10.1093/joccuh/uiae053

89. Bhattacharya, Y. (2019) Hazard and near-miss reporting - safety through numbers?. *Journal of Maritime Research*. https://www.scopus.com/pages/publications/85116431106

90. Khatsuriya, J.K., Gaur, A. (2025) Perceived effectiveness of near-miss reporting on proactive safety performance in construction: A mixed-methods study from India. *International Journal of Occupational Safety and Health*. https://doi.org/10.3126/ijosh.v15i3.80888

91. Brazier, A.J. (1994) A summary of incident reporting in the process industry. *Journal of Loss Prevention in the Process Industries*. https://doi.org/10.1016/0950-4230(94)80072-3

92. Sendelbach, S., Funk, M. (2013) Alarm fatigue: A patient safety concern. *AACN Advanced Critical Care*. https://doi.org/10.1097/NCI.0b013e3182a903f9

93. Shi, W., Cao, J., Zhang, Q., (...), Xu, L. (2016) Edge Computing: Vision and Challenges. *IEEE Internet of Things Journal*. https://doi.org/10.1109/JIOT.2016.2579198

94. Zheng, R., Ng, S.T., Shao, Y., (...), Xing, J. (2025) Leveraging digital twin for healthcare emergency management system: Recent advances, critical challenges, and future directions. *Reliability Engineering and System Safety*. https://doi.org/10.1016/j.ress.2025.111079

95. Ferreira, Daniel Brantes, Gromova, Elizaveta A (2026) The Bossware Era and the e-Panopticon: Current Technologies and Legal Challenges. *Industrial Law Journal*. https://doi.org/10.1093/indlaw/dwaf006

96. Putz, S., Rick, V., Mertens, A., Nitsch, V. (2022) Using IoT devices for sensor-based monitoring of employees' mental workload: Investigating managers' expectations and concerns. *Applied Ergonomics*. https://doi.org/10.1016/j.apergo.2022.103739

97. Barlybayev, A., Milosz, M., Amangeldy, N., (...), Lamasheva, Z. (2026) Systematic Review of Computer-Vision Technologies for Personal Protective Equipment Compliance Monitoring. *Computers*. https://doi.org/10.3390/computers15060388

98. Setiawan, D.L., Muchtar, K., Muchallil, S. (2026) Real-Time Safety Helmet Compliance Monitoring Using AI-Driven Multi-CCTV Video Analytics. *Proceedings of the 2026 IEEE International Conference on Instrumentation, Networks, Future Sustainability, Information Theory, Transportation, and Systems, INFINITYS 2026*. https://doi.org/10.1109/INFINITYS69204.2026.11666189

99. Evans, E.W., Samuel, E.J., Mayho, S., (...), Redmond, E.C. (2025) Assessing hand hygiene in food manufacturing: The feasibility of CCTV surveillance. *Food Control*. https://doi.org/10.1016/j.foodcont.2025.111282

100. Ramadan, M.N.A., Ali, M.A.H., Khoo, S.Y., AlKhedher, M. (2025) SecureIoT-FL: A Federated Learning Framework for Privacy-Preserving Real-Time Environmental Monitoring in Industrial IoT Applications. *Alexandria Engineering Journal*. https://doi.org/10.1016/j.aej.2024.11.069

101. McMahan, H., Moore, E., Ramage, D., (...), Agüera y Arcas, B. (2017) Communication-efficient learning of deep networks from decentralized data. *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics, AISTATS 2017*. https://www.scopus.com/pages/publications/85083937116

102. Chen, M. (2026) Real-Time Compliance Monitoring Engine Based on eBPF: Privacy-Enabled Data Tracking for Edge Computing. *Procedia Computer Science*. https://doi.org/10.1016/j.procs.2026.04.210

103. Gnanasekaran, V., Bartnes, M., Heegaard, P.E. (2026) A framework for onboarding security operations centers in industrial environments. *Information and Computer Security*. https://doi.org/10.1108/ICS-03-2026-0145

104. Juarez Juarez, M.G.J., Giret, A., Botti, V. (2025) Semantic and modular orchestration of AI-driven digital twins for industrial interoperability and optimization. *Journal of Industrial Information Integration*. https://doi.org/10.1016/j.jii.2025.100959

105. Bongomin, O., Mwape, M.C., Mpofu, N.S., (...), Ngulube, G. (2025) Digital twin technology advancing industry 4.0 and industry 5.0 across sectors. *Results in Engineering*. https://doi.org/10.1016/j.rineng.2025.105583

106. Maranco, M., Sivakumar, M., Nagappan, N., (...), Nidhya, R. (2025) Digital Twin-Enabled Smart Manufacturing: Challenges and Future Directions. *Artificial Intelligence-Enabled Digital Twin for Smart Manufacturing*. https://doi.org/10.1002/9781394303601.ch21

107. Ferko, E., Bucaioni, A., Pelliccione, P., Behnam, M. (2023) Standardisation in Digital Twin Architectures in Manufacturing. *Proceedings - IEEE 20th International Conference on Software Architecture, ICSA 2023*. https://doi.org/10.1109/ICSA56044.2023.00015

108. Iñigo, M.A., Porto, A., Kremer, B., (...), Cuenca, J. (2020) Towards an Asset Administration Shell scenario: A use case for interoperability and standardization in Industry 4.0. *Proceedings of IEEE/IFIP Network Operations and Management Symposium 2020: Management in the Age of Softwarization and Artificial Intelligence, NOMS 2020*. https://doi.org/10.1109/NOMS47738.2020.9110410

109. Diaz, E. (2006) Pick the right camera interface. *Sensors (Peterborough, NH)*. https://www.scopus.com/pages/publications/54249090238

110. Camillo, J. (2013) Choosing a vision interface standard. *Assembly*. https://www.scopus.com/pages/publications/84878421604

111. Hopfner, T. (2024) Greater Interoperability through Unified Machine Vision Standards. *ZWF Zeitschrift fuer Wirtschaftlichen Fabrikbetrieb*. https://doi.org/10.1515/zwf-2024-1126

112. Okhuijsen, B. (2022) Combining the Process and Maintenance Digital Twin to Create an Autonomous Production Platform. *Proceedings of the Annual Offshore Technology Conference*. https://doi.org/10.4043/31773-MS

113. Varghese, Allen, Starly, Binil, Iquebal, Ashif (2026) Vector Quantization of Complex Sensor Data in Limited Network Bandwidth and Edge Hardware Constrained Digital Twins. *Manufacturing Letters*. https://doi.org/10.1016/j.mfglet.2026.08.060